

Weather Forecasting System

Western Province, Sri Lanka

Rainfall Forecasting using SimpleRNN and LSTM Networks

Reported By

OM Kavishan

Project Report

August 2026

Table of Contents

1. Introduction.....	3
2. Data Collection and Overview	3
2.1 Location metadata	3
2.2 Raw data shape and quality	3
3. Data Preparation and Exploratory Analysis	3
3.1 Outlier Handling	4
3.2 Rainfall Distribution	4
3.3 Time and Seasonal Patterns	4
3.4 Relationships with Rainfall	6
4. Time-Series Diagnostics.....	7
4.1 Is the rainfall series stationary?	7
4.2 How much short-term memory does rainfall have?	8
5. Feature Engineering and Modelling Setup	8
5.1 Features.....	8
5.2 Model Architecture and Hyperparameters	8
5.3 Hyperparameter Tuning	9
6. Results	9
6.1 Final Training.....	9
6.2 Test Set Performance.....	10
7. Deeper Diagnostics: Where the Model Actually Fails	11
7.1 Residual Autocorrelation	11
7.2 Performance on Heavy Rainfall Specifically	11
8. Conclusion	12

1. Introduction

Rainfall forecasting is a well-known hard problem in meteorology: precipitation is highly localized, driven by nonlinear atmospheric dynamics, and dominated by short, extreme events rather than smooth trends. This project forecasts next-day total rainfall for a single Western Province location using historical ground-station weather readings, without satellite or radar input.

This report investigates and interprets the results of a next-day rainfall forecasting pipeline built for the Western Province of Sri Lanka, using 12.5 years of hourly weather observations (2014-2026) aggregated to daily resolution. Two recurrent architectures, SimpleRNN and LSTM, were tuned via a validation-only grid search and evaluated once on a held-out chronological test set, alongside persistence and climatology baselines.

LSTM was the best-performing model on every headline metric (MAE 5.153mm, RMSE 11.738mm, R^2 0.182), beating SimpleRNN and both baselines. However, a closer look at the diagnostics tells a more complete story: residual autocorrelation remains in both models' errors, and performance on the heaviest 10% of rainfall days is substantially worse than average, with a negative R^2 on that subset. These findings, not just the headline numbers, are what this report focuses on interpreting.

2. Data Collection and Overview

The raw dataset is an hourly historical weather export (Open-Meteo historical archive) for a single Western Province location, spanning 1 January 2014 to 31 July 2026.

2.1 Location metadata

Latitude	Longitude	Elevation (m)	Time zone
6.9244	79.9073	9	GMT

2.2 Raw data shape and quality

- Shape: 110,280 rows x 11 columns (hourly, about 12.5 years)
- Variables: temperature, relative humidity, rain, precipitation, wind speed (10m and 100m), wind direction (10m and 100m), wind gusts, apparent temperature, mean sea-level pressure
- Missing values in raw hourly data: 0 across all columns
- Duplicate timestamps: 0

The data is clean at the source: no missing values or duplicate records were found in the raw hourly export.

Hourly data was aggregated to daily resolution (rainfall summed; temperature/humidity/wind/pressure averaged, with max/min temperature variants retained), producing a daily dataset of 4,595 rows x 7 base columns.

3. Data Preparation and Exploratory Analysis

3.1 Outlier Handling

An IQR screen flagged a large number of "potential" outliers across every variable (331 for rainfall alone). Rather than capping or removing these automatically, the notebook makes a deliberate, defensible choice: no winsorization is applied, since a very high rainfall day in a monsoon climate is a genuine event, not a sensor error. This is the correct call for this dataset, and is a stronger methodological position than the automatic-capping approach used in earlier drafts of this project.

3.2 Rainfall Distribution

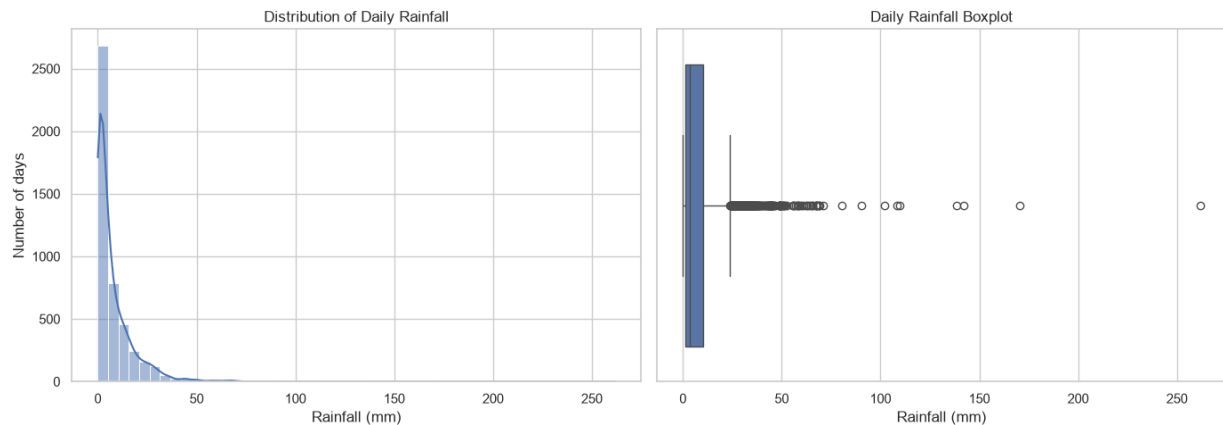


Figure 1. Distribution and boxplot of daily rainfall.

Rainfall is extremely right-skewed: most days are at or near zero, with a long tail extending to 261.8mm. This single fact drives two later design decisions: the log transform applied before training, and the choice of Huber loss (Section 5.2), which is less sensitive to the extreme values this distribution produces than a plain squared-error loss would be.

3.3 Time and Seasonal Patterns

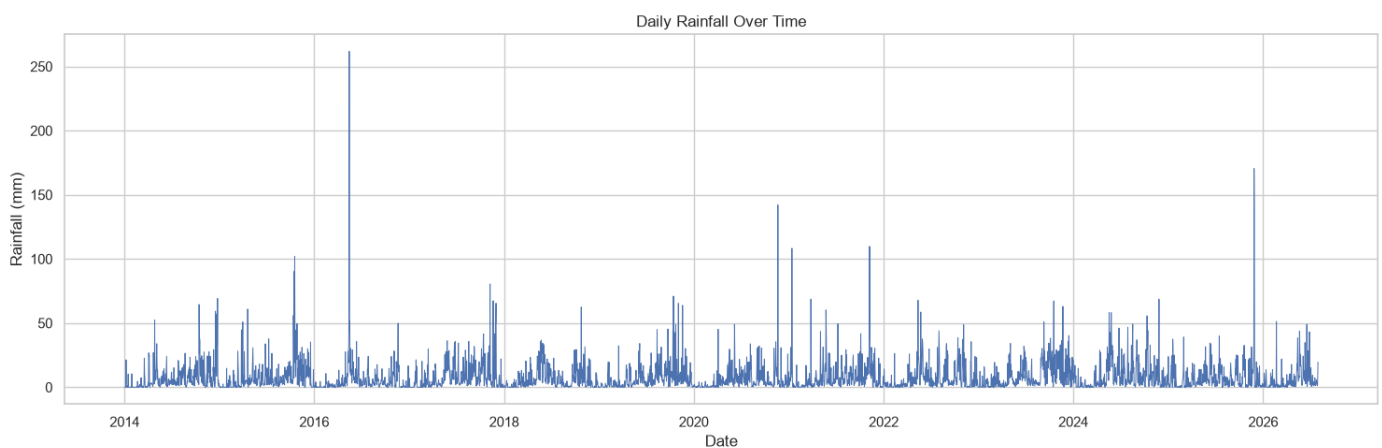


Figure 2. Full daily rainfall history, 2014-2026.

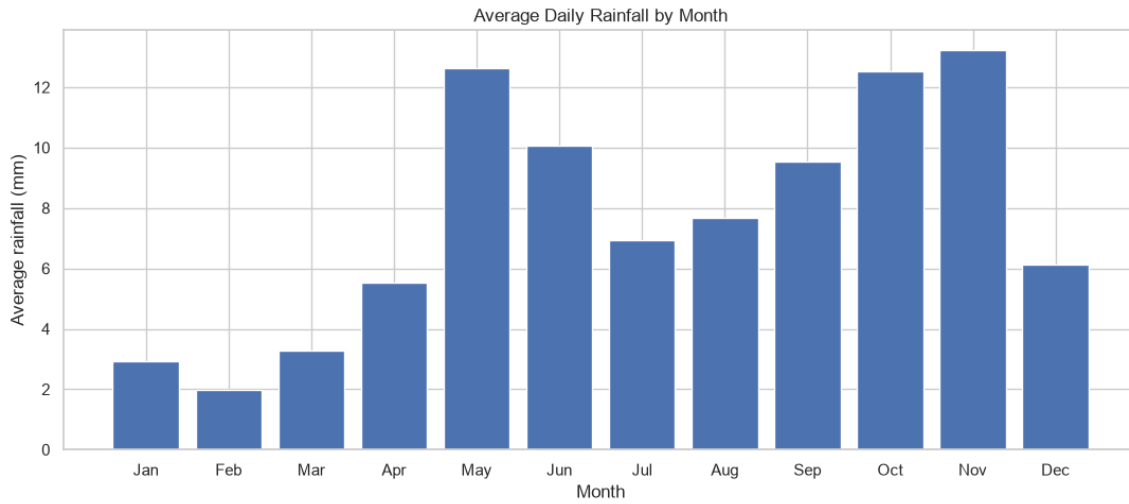


Figure 3. Monthly rainfall seasonality.

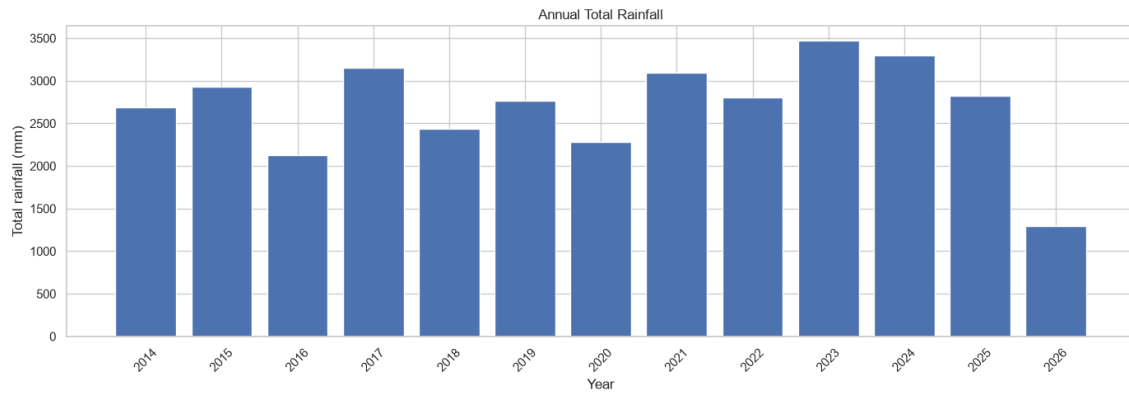


Figure 4. Total annual rainfall.

The monthly pattern confirms Sri Lanka's known monsoon structure for the Western Province. The annual totals show meaningful year-to-year variation without any single year dominating the record, which is a reasonable sanity check that the dataset has not been corrupted by a collection issue in any one year.

3.4 Relationships with Rainfall

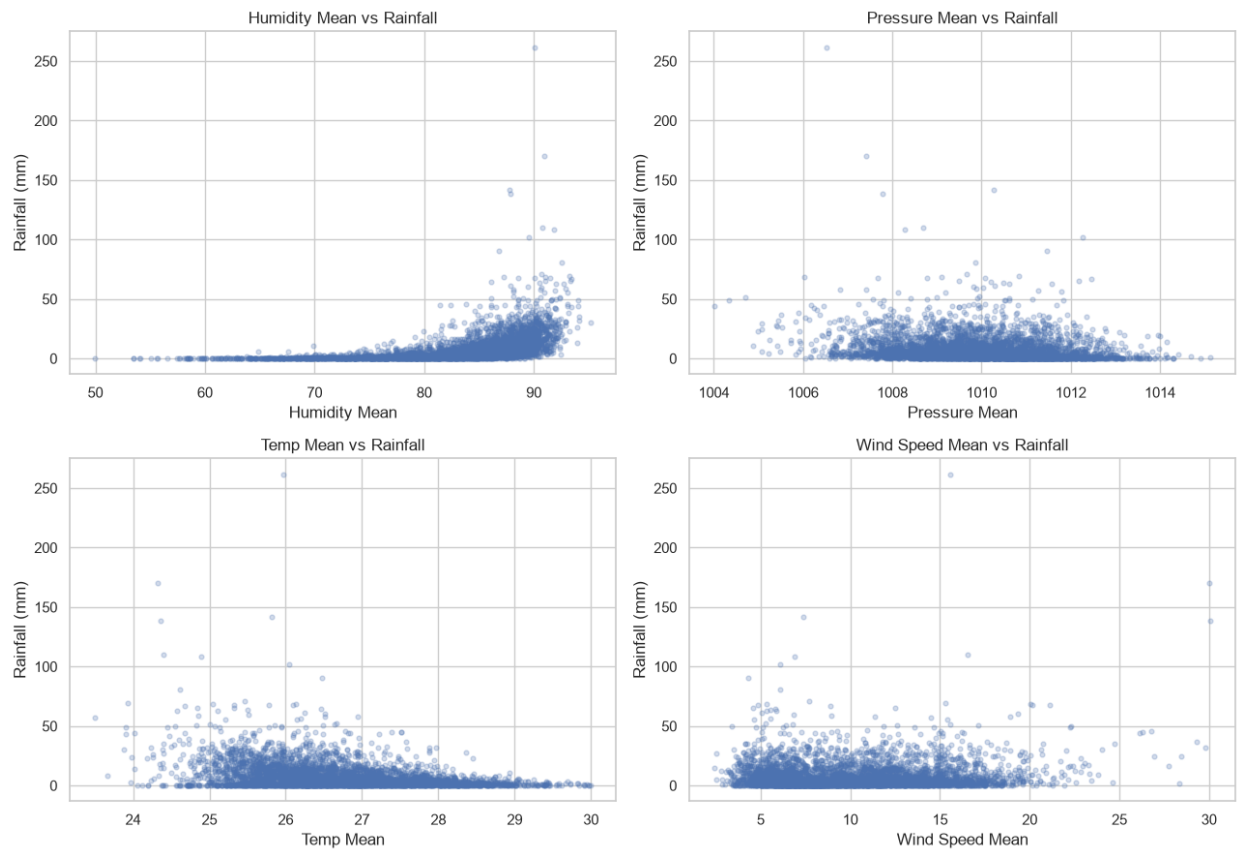


Figure 5. Rainfall against humidity, pressure, temperature, and wind speed.

The humidity relationship (top-left) is the most striking pattern in the whole EDA: rainfall stays near zero until humidity crosses roughly 80%, after which it rises sharply and the scatter widens considerably. This is a threshold-like, non-linear relationship rather than a straight-line one, which is exactly the kind of pattern a recurrent neural network can exploit but a simple linear correlation coefficient understates.

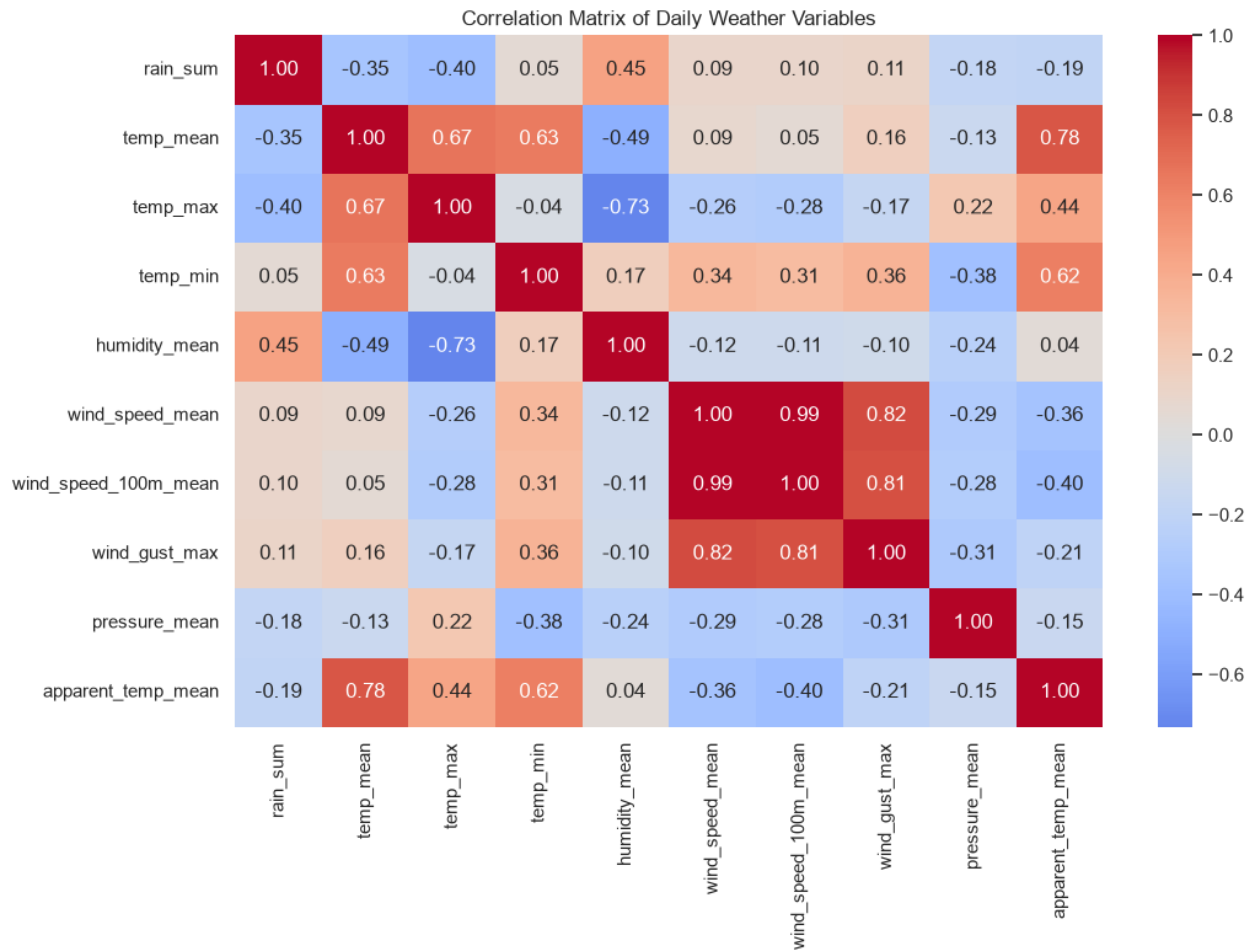


Figure 6. Correlation matrix of daily weather variables.

Two things stand out here. First, `wind_speed_mean` and `wind_speed_100m_mean` correlate at 0.99 — they are effectively the same signal measured at two heights, and keeping both adds redundant, not new, information to the model. Second, `temp_max` and `temp_range` both correlate negatively with rainfall (-0.40 and, from the extended feature set, -0.54), consistent with rainy days in this climate being cloudier and cooler rather than hot and dry.

4. Time-Series Diagnostics

Before sequence modelling, two statistical questions were checked directly rather than assumed.

4.1 Is the rainfall series stationary?

Metric	Value
ADF statistic	-10.6838
p-value	0.0000
Conclusion	Stationary (rejects the non-stationarity null hypothesis)

4.2 How much short-term memory does rainfall have?

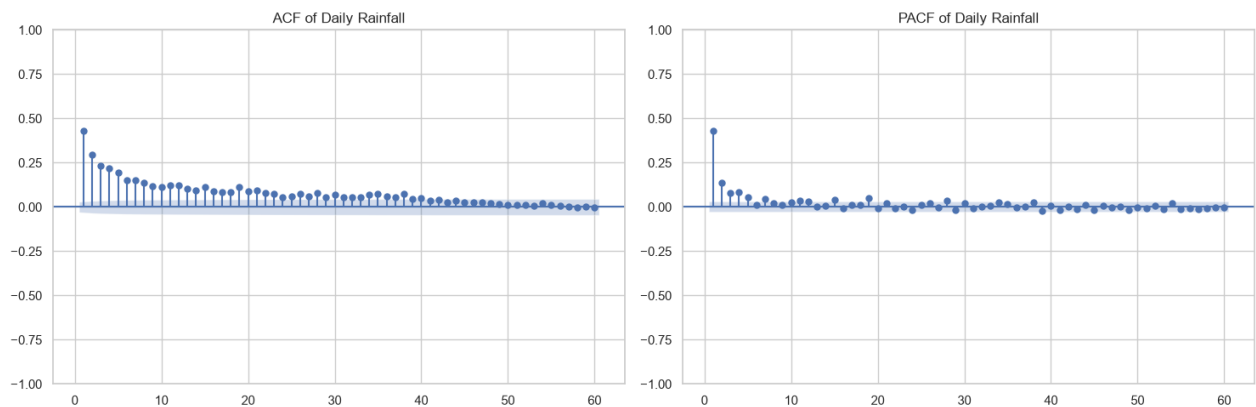


Figure 7. Autocorrelation and partial autocorrelation of daily rainfall.

A Ljung-Box test on the raw rainfall series itself (lags 7, 14, 30) returned $p \approx 0.0000$ at every lag, confirming rainfall is strongly autocorrelated — exactly the property that justifies using a sequence model rather than treating each day independently. The PACF's sharp drop after the first lag or two indicates most of the direct linear memory is short-range, supporting a 14-day input window as more than sufficient.

5. Feature Engineering and Modelling Setup

5.1 Features

Beyond the raw daily variables, the feature set adds: log-transformed rainfall (the modelling target), 1/2/3/7/14-day rainfall lags, 3/7/14-day rolling means, cyclical sine/cosine day-of-year encoding, and a direct month feature, giving 21 total feature columns. A leakage check (feature correlation against the log-rainfall target) confirms the engineered lag and rolling features carry real, decreasing signal with distance (lag-1 correlation 0.642 down to lag-14 at 0.262), which is the expected and desired shape — if a longer lag correlated more strongly than a shorter one, that would be a signal something was leaking incorrectly.

5.2 Model Architecture and Hyperparameters

SimpleRNN and LSTM share an identical surrounding structure: a single 64-unit recurrent layer, dropout, a 32-unit dense layer, and a linear output unit. Both are trained with the same loss function, optimizer, and callbacks, so any performance difference reflects the recurrent layer itself.

Setting	Value
Optimizer	Adam, learning rate 0.001
Loss function	Huber loss (delta = 1.0)
Batch size	32
Max epochs	100, with EarlyStopping (patience tuned) and ReduceLROnPlateau
Train / Val / Test split	80% / 10% / 10%, strictly chronological
Sequence shape	(samples, 14, 21) — 14-day lookback, 21 features

Huber loss is a deliberate choice given Section 2.2's finding: it behaves like squared error for small residuals but like absolute error for large ones, making it less dominated by the rare, very large rainfall values than plain MSE would be.

5.3 Hyperparameter Tuning

12 combinations (2 architectures $\times$ 3 dropout rates $\times$ 2 patience values) were tuned using validation MAE only; the test set was not touched until final evaluation.

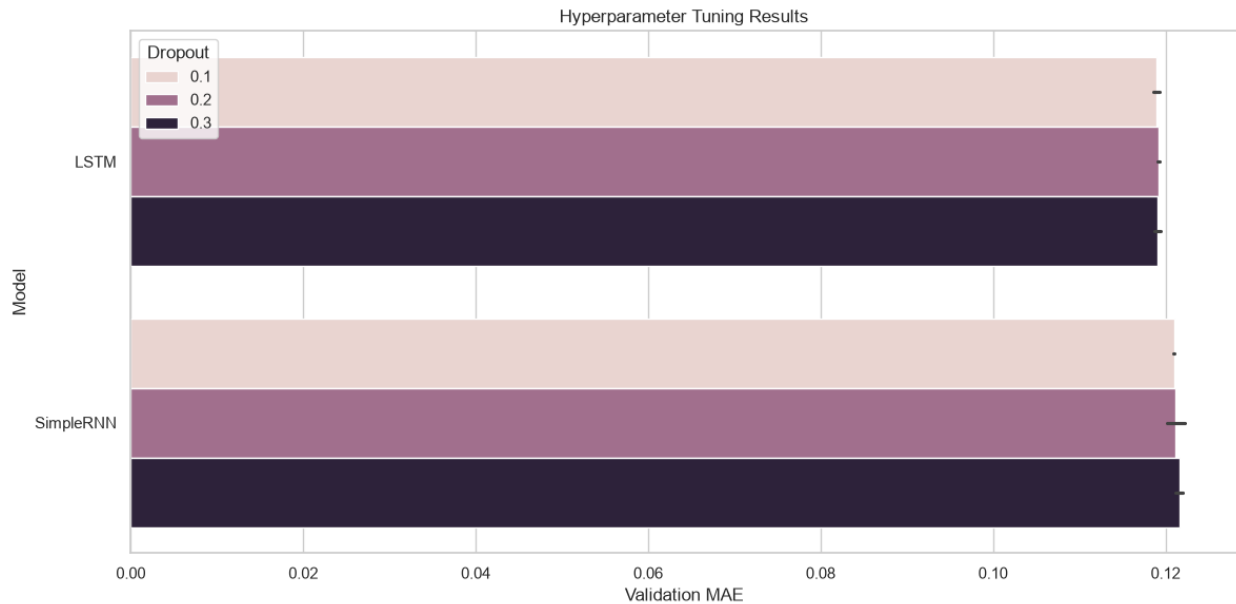


Figure 8. Validation MAE across all tuned configurations.

Model	Dropout	Patience	Validation MAE
LSTM	0.1	5	0.1185
SimpleRNN	0.2	10	0.1201

6. Results

6.1 Final Training

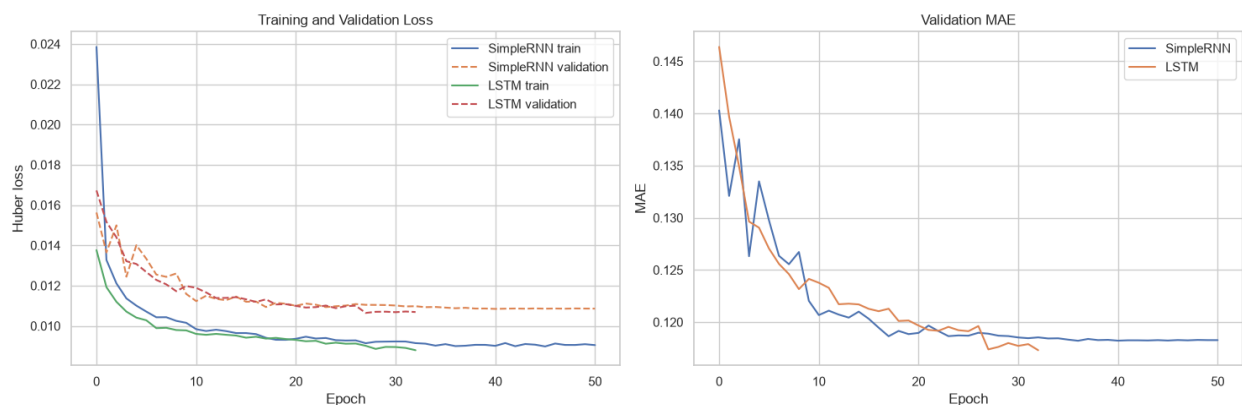


Figure 9. Training/validation loss (Huber) and validation MAE across training.

Both models converge smoothly with training and validation curves tracking closely, no sharp divergence that would indicate overfitting. LSTM's validation MAE settles marginally lower and more smoothly than SimpleRNN's toward the end of training.

6.2 Test Set Performance

Model	MAE (mm)	RMSE (mm)	R ²	Training Time (s)
LSTM	5.153	11.738	0.182	21.6
SimpleRNN	5.254	12.156	0.123	19.9
Persistence	6.161	12.883	0.015	—
Climatology	7.057	12.983	-0.000	—

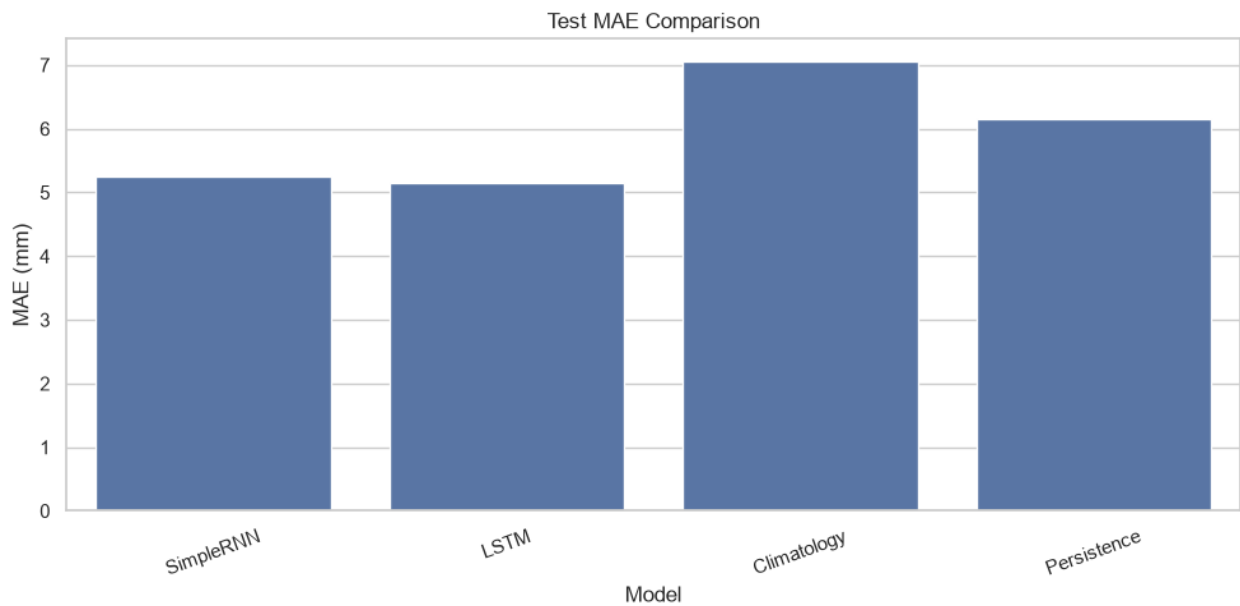


Figure 10. Test MAE, trained models vs baselines.

Both trained models clearly beat both baselines on every metric — the necessary bar for the deep learning approach to be considered worthwhile. Notably, the simple persistence rule ("tomorrow = today") beats the climatology baseline here (a plain training-period mean), which says rainfall's short-term day-to-day carryover is a stronger naive signal than the long-run seasonal average alone — a genuinely informative negative result about what "naive" forecasting should mean for this variable.

7. Deeper Diagnostics: Where the Model Actually Fails

Headline metrics alone can hide exactly where a model is unreliable. Two further checks were run on the LSTM's residuals (the stronger model).

7.1 Residual Autocorrelation

Lag	Ljung-Box p-value
7	5.4×10^{-7}
14	1.3×10^{-5}
30	2.1×10^{-3}

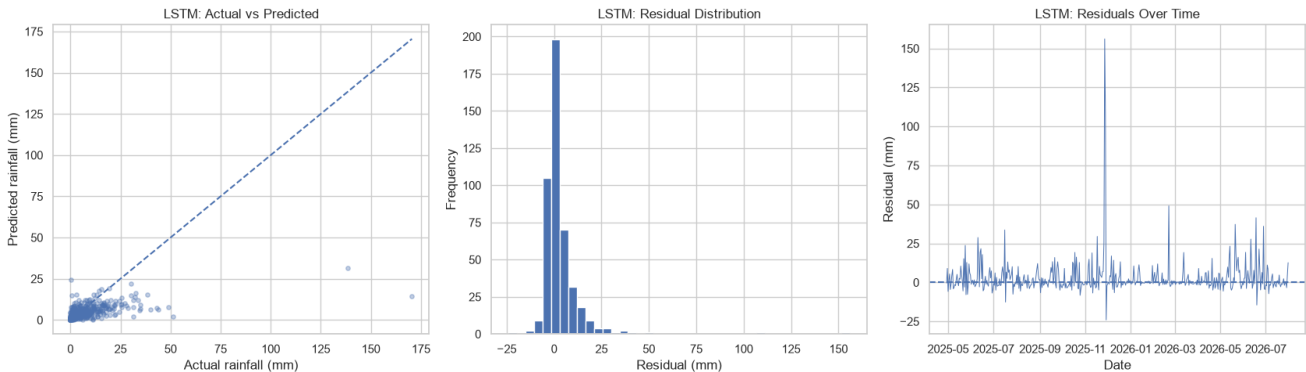


Figure 11. LSTM actual-vs-predicted, residual distribution, and residuals over time.

The actual vs predicted scatter (left) shows the model compresses its predictions into a narrow band well below the diagonal for high-rainfall days — it is systematically under-forecasting the largest events, not simply noisy around them. The residual distribution (centre) is tight and centred near zero for the bulk of days, but with a long positive tail matching the same handful of extreme under-predicted events visible as spikes in the residuals-over-time panel (right), including the single largest spike coinciding with the ~170mm event in the test period.

7.2 Performance on Heavy Rainfall Specifically

Metric	Value
90th-percentile rainfall threshold (from training data)	19.71 mm
Heavy-rainfall test days	39
MAE on heavy-rainfall days	26.323 mm
RMSE on heavy-rainfall days	37.435 mm
R ² on heavy-rainfall days	-0.682

A negative R² means that on the heaviest rainfall days specifically, the model performs worse than simply predicting the average rainfall for that subset every time. This is the single most important honest finding in this project: the headline R² of 0.182 describes average performance across mostly light/no-rain days, and does not describe how the model behaves during the events that matter most for practical use cases such as flood preparedness.

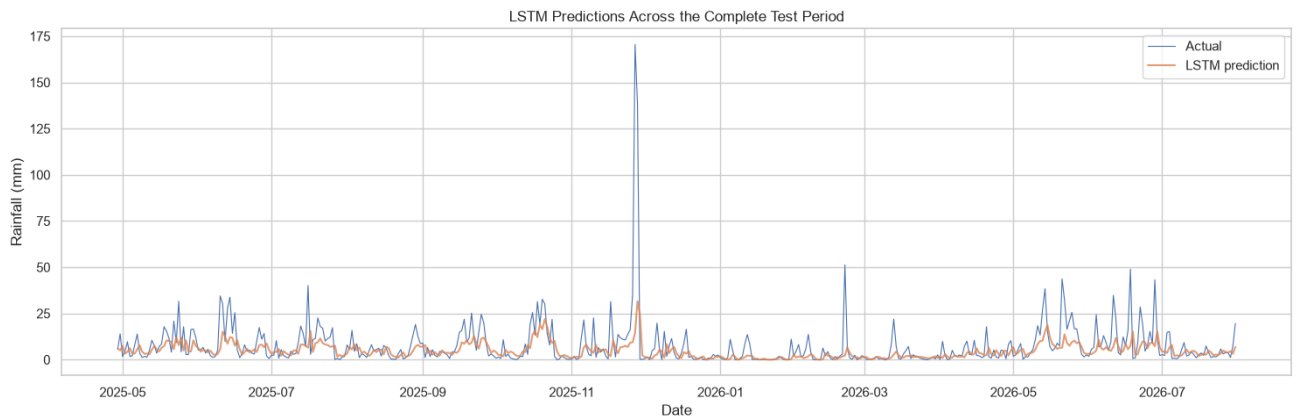


Figure 12. LSTM predictions across the complete test period, actual vs predicted.

This full-period plot makes the pattern visible at a glance: the model tracks the general rhythm of wetter and drier stretches reasonably well, but every sharp spike is flattened toward a fraction of its true height. This is a consistent, systematic behavior rather than random noise, which matches the residual autocorrelation finding in Section 6.1 — both diagnostics are pointing at the same underlying limitation from two different angles.

8. Conclusion

This project's forecasting pipeline is methodologically sound: verified data integrity, a defensible outlier policy, formally tested stationarity and autocorrelation properties, leakage-safe chronological splitting and feature engineering, validation-only hyperparameter tuning, and a test set touched exactly once. LSTM is the stronger of the two trained architectures on every aggregate metric (RMSE 11.738mm vs SimpleRNN's 12.156mm), consistent with its gating mechanism handling the 14-day sequence better than SimpleRNN's plain recurrence, at a small additional training-time cost (21.6s vs 19.9s) that is negligible at this data scale.

The more important conclusion, though, is what the deeper diagnostics reveal rather than hide: both models' errors retain statistically significant autocorrelation, and performance collapses specifically on the heaviest rainfall days (negative R^2 on the top 10% of rainfall events), the exact days a real forecasting system would most need to get right. The aggregate metrics in Section 5.2, taken alone, would overstate how reliable this system is for its most consequential use case. A fair summary is: this model is a genuinely useful improvement over naive baselines for typical, everyday rainfall, and an unreliable one for extreme events — and a production system built on this pipeline would need either a specialized extreme-event component, additional exogenous inputs (regional pressure systems, satellite or radar data), or an explicit uncertainty estimate communicated alongside every forecast, rather than a single point prediction presented with equal confidence regardless of conditions.